

Commerce XI Maths App

Theory (Pages 20–30)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 2 - Functions | Total questions: 41

Question 1 - Theory P20

A function f from set A to set B associates each element x in A with how many elements y in B ?

Options

- A. Exactly one
- B. At least two
- C. No element
- D. Any number, including none

Correct answer: Exactly one

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is Exactly one.

Question 2 - Theory P20

If $f: A \rightarrow B$ is a function, set A is called the

Options

- A. domain of f
- B. range of f
- C. co-domain only
- D. image of f

Correct answer: domain of f

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is domain of f .

Question 3 - Theory P20

If $f: A \rightarrow B$ and $y = f(x)$, then y is called the

Options

- A. image of x under f
- B. pre-image of x
- C. domain of f
- D. inverse of x

Correct answer: image of x under f

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is image of x under f .

Question 4 - Theory P20

If $f: A \rightarrow B$ and $y = f(x)$, then x is called the

Options

- A. pre-image of y under f
- B. image of y
- C. co-domain of y
- D. range of y

Correct answer: pre-image of y under f

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is pre-image of y under f .

Question 5 - Theory P20

A relation from A to B is not a function if an element of A

Options

- A. has no image or has more than one image in B
- B. has exactly one image in B
- C. has the same name as an element of B
- D. is joined by a curved arrow

Correct answer: has no image or has more than one image in B

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is has no image or has more than one image in B .

Question 6 - Theory P20-21

A function $f: A \rightarrow B$ is one-one if

Options

- A. distinct elements in A have distinct images in B
- B. every element of B has two pre-images
- C. all elements of A have the same image
- D. A and B contain the same elements

Correct answer: distinct elements in A have distinct images in B

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is distinct elements in A have distinct images in B.

Question 7 - Theory P21

A function $f: A \rightarrow B$ is onto if

Options

- A. every element of B is an image of some element of A
- B. every element of A has two images
- C. some element of B has no pre-image
- D. $f(A)$ is a proper subset of B

Correct answer: every element of B is an image of some element of A

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is every element of B is an image of some element of A.

Question 8 - Theory P21

For an onto function $f: A \rightarrow B$, which equality holds?

Options

- A. $f(A) = B$
- B. $f(A) = A$
- C. $f(A) = \emptyset$
- D. $A \cap B = \emptyset$

Correct answer: $f(A) = B$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(A) = B$.

Question 9 - Theory P21

A one-one and onto function is also called

Options

- A. bijective
- B. constant
- C. into

D. step

Correct answer: bijective

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is bijective.

Question 10 - Theory P21

A function is into when

Options

- A. at least one element of the co-domain has no pre-image
- B. every element of the co-domain has a pre-image
- C. every input has two outputs
- D. its range equals its co-domain

Correct answer: at least one element of the co-domain has no pre-image

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is at least one element of the co-domain has no pre-image.

Question 11 - Theory P21-22

In the ordered-pair representation of a function, the domain is the set of

Options

- A. first components
- B. second components
- C. all components
- D. products of components

Correct answer: first components

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is first components.

Question 12 - Theory P21-22

In the ordered-pair representation of a function, the range is the set of

Options

- A. second components
- B. first components

- C. ordered pairs themselves
- D. unused co-domain elements

Correct answer: second components

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is second components.

Question 13 - Theory P22

For $y = f(x) = 2x + 1$, $f(x)$ is read as

Options

- A. f of x
- B. f multiplied by x
- C. inverse of x
- D. x of f

Correct answer: f of x

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is f of x .

Question 14 - Theory P22

In graphical form, the domain is read from the extent of the graph on the

Options

- A. x-axis
- B. y-axis
- C. line $y = x$
- D. origin only

Correct answer: x-axis

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is x-axis.

Question 15 - Theory P22

In graphical form, the range is read from the extent of the graph on the

Options

- A. y-axis

- B. x-axis
- C. line $y = x$
- D. origin only

Correct answer: y-axis

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is y-axis.

Question 16 - Solved Example P22

For $f(x) = 2x^2 - 3x + 4$, find $f(7)$.

Correct answer: 81

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives 81. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 17 - Solved Example P22

From the textbook graph of $y = g(x)$, $g(-4)$ equals

Options

- A. 0
- B. -4
- C. 3
- D. -5

Correct answer: 0

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives 0. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 18 - Solved Example P22

From the textbook graph of $y = g(x)$, $g(3)$ equals

Options

- A. -5
- B. 0
- C. 3
- D. 5

Correct answer: -5

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives -5 . The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 19 - Solved Example P22-23

If $f(x) = 3x^2 - x$ and $f(m) = 4$, then m equals

Options

- A. $4/3$ or -1
- B. 1 or $-4/3$
- C. 4 or -1
- D. $1/3$ or -4

Correct answer: $4/3$ or -1

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives $4/3$ or -1 . The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 20 - Theory P23

A constant function $f: \mathbb{R} \rightarrow \mathbb{R}$ has the form

Options

- A. $f(x) = k$
- B. $f(x) = x$
- C. $f(x) = ax + b$, $a \neq 0$
- D. $f(x) = 1/x$

Correct answer: $f(x) = k$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(x) = k$.

Question 21 - Theory P23

For the constant function $f(x) = 3$, its range is

Options

- A. $\{3\}$
- B. $\mathbb{R}$
- C. $[0, \infty)$
- D. $\mathbb{R} - \{0\}$

Correct answer: {3}

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is {3}.

Question 22 - Theory P24

The identity function on $\mathbb{R}$ is defined by

Options

A. $f(x) = x$

B. $f(x) = 1$

C. $f(x) = x^2$

D. $f(x) = |x|$

Correct answer: $f(x) = x$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(x) = x$.

Question 23 - Theory P24

The graph of the linear function $f(x) = ax + b$ has slope and y-intercept respectively

Options

A. a and b

B. b and a

C. $-b/a$ and a

D. a and $-b/a$

Correct answer: a and b

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is a and b.

Question 24 - Theory P24

The graph of $f(x) = ax + b$ is increasing when

Options

A. $a > 0$

B. $a < 0$

C. $a = 0$ only

D. $b < 0$

Correct answer: $a > 0$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $a > 0$.

Question 25 - Theory P24

For a quadratic function whose graph has minimum value k , the range is

Options

- A. $[k, \infty)$
- B. $(-\infty, k]$
- C. $\mathbb{R} - \{k\}$
- D. $\{k\}$

Correct answer: $[k, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $[k, \infty)$.

Question 26 - Theory P24-25

The domain and range of the square function $f(x) = x^2$ are

Options

- A. Domain $\mathbb{R}$, Range $[0, \infty)$
- B. Domain $[0, \infty)$, Range $\mathbb{R}$
- C. Domain $\mathbb{R}$, Range $\mathbb{R}$
- D. Domain $\mathbb{R} - \{0\}$, Range $\mathbb{R} - \{0\}$

Correct answer: Domain $\mathbb{R}$, Range $[0, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is Domain $\mathbb{R}$, Range $[0, \infty)$.

Question 27 - Theory P25

The domain and range of the cube function $f(x) = x^3$ are

Options

- A. Domain $\mathbb{R}$, Range $\mathbb{R}$
- B. Domain $\mathbb{R}$, Range $[0, \infty)$
- C. Domain $\mathbb{R} - \{0\}$, Range $\mathbb{R}$

D. Domain $[0, \infty)$, Range $\mathbb{R}$

Correct answer: Domain $\mathbb{R}$, Range $\mathbb{R}$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is Domain $\mathbb{R}$, Range $\mathbb{R}$.

Question 28 - Theory P25

A rational function has the form

Options

A. $f(x) = p(x)/q(x)$, $q(x) \neq 0$

B. $f(x) = ax + b$ only

C. $f(x) = ax^n$ only

D. $f(x) = \log_a x$ only

Correct answer: $f(x) = p(x)/q(x)$, $q(x) \neq 0$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(x) = p(x)/q(x)$, $q(x) \neq 0$.

Question 29 - Theory P25

For $f(x) = 1/x$, the domain and range are

Options

A. $\mathbb{R} - \{0\}$ and $\mathbb{R} - \{0\}$

B. $\mathbb{R}$ and $\mathbb{R}$

C. $[0, \infty)$ and $\mathbb{R}$

D. $\mathbb{R}$ and $\{0\}$

Correct answer: $\mathbb{R} - \{0\}$ and $\mathbb{R} - \{0\}$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $\mathbb{R} - \{0\}$ and $\mathbb{R} - \{0\}$.

Question 30 - Theory P26

For an exponential function $f(x) = a^x$, the required conditions on a are

Options

A. $a > 0$ and $a \neq 1$

B. $a < 0$

C. $a = 1$

D. $a = 0$

Correct answer: $a > 0$ and $a \neq 1$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $a > 0$ and $a \neq 1$.

Question 31 - Theory P26

The domain and range of $f(x) = a^x$ are

Options

A. $\mathbb{R}$ and $(0, \infty)$

B. $(0, \infty)$ and $\mathbb{R}$

C. $\mathbb{R}$ and $\mathbb{R}$

D. $\mathbb{R} - \{0\}$ and $\mathbb{R} - \{0\}$

Correct answer: $\mathbb{R}$ and $(0, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $\mathbb{R}$ and $(0, \infty)$.

Question 32 - Theory P26

The logarithmic statement $y = \log_a x$ is equivalent to

Options

A. $a^y = x$

B. $x^y = a$

C. $a^x = y$

D. $y^a = x$

Correct answer: $a^y = x$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $a^y = x$.

Question 33 - Theory P26

The domain of the logarithmic function $f(x) = \log_a x$ is

Options

- A. $(0, \infty)$
- B. $\mathbb{R}$
- C. $[0, \infty)$
- D. $\mathbb{R} - \{0\}$

Correct answer: $(0, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $(0, \infty)$.

Question 34 - Theory P27

For functions f and g , $(f + g)(x)$ equals

Options

- A. $f(x) + g(x)$
- B. $f(g(x))$
- C. $f(x)g(x)$
- D. $f(x)/g(x)$

Correct answer: $f(x) + g(x)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(x) + g(x)$.

Question 35 - Theory P27-29

If $f: A \rightarrow B$ and $g: B \rightarrow C$, then $(g \circ f)(x)$ equals

Options

- A. $g[f(x)]$
- B. $f[g(x)]$
- C. $g(x) + f(x)$
- D. $g(x)f(x)$

Correct answer: $g[f(x)]$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $g[f(x)]$.

Question 36 - Theory P27

For functions f and g , $(f/g)(x)$ is defined as $f(x)/g(x)$, provided

Options

- A. $g(x) \neq 0$
- B. $f(x) = 0$
- C. $g(x) = 0$
- D. $f(x) = g(x)$

Correct answer: $g(x) \neq 0$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $g(x) \neq 0$.

Question 37 - Theory P28

If f is one-one and onto and $f(x) = y$, then

Options

- A. $f^{-1}(y) = x$
- B. $f^{-1}(x) = y$ always
- C. $f(y) = x$ always
- D. f^{-1} does not exist

Correct answer: $f^{-1}(y) = x$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f^{-1}(y) = x$.

Question 38 - Theory P28

If f and g are inverse functions, then

Options

- A. $f[g(x)] = x$ and $g[f(x)] = x$
- B. $f[g(x)] = 0$
- C. $f(x) + g(x) = x$
- D. $f(x) = g(x)$ for every x

Correct answer: $f[g(x)] = x$ and $g[f(x)] = x$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f[g(x)] = x$ and $g[f(x)] = x$.

Question 39 - Theory P29-30

The range of the signum function $\text{sgn}(x)$ is

Options

- A. $\{-1, 0, 1\}$
- B. $\mathbb{R}$
- C. $[0, \infty)$
- D. Set of integers

Correct answer: $\{-1, 0, 1\}$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $\{-1, 0, 1\}$.

Question 40 - Theory P30

For the absolute value function $f(x) = |x|$, the range is

Options

- A. $[0, \infty)$
- B. $\mathbb{R}$
- C. $(-\infty, 0]$
- D. $\{-1, 0, 1\}$

Correct answer: $[0, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $[0, \infty)$.

Question 41 - Theory P30

For the greatest integer function $f(x) = [x]$, the range is

Options

- A. $\mathbb{I}$, the set of integers
- B. $\mathbb{R}$
- C. $[0, \infty)$
- D. $\{-1, 0, 1\}$

Correct answer: $\mathbb{I}$, the set of integers

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $\mathbb{I}$, the set of integers.